RAG Learning

Topic 10 — Space

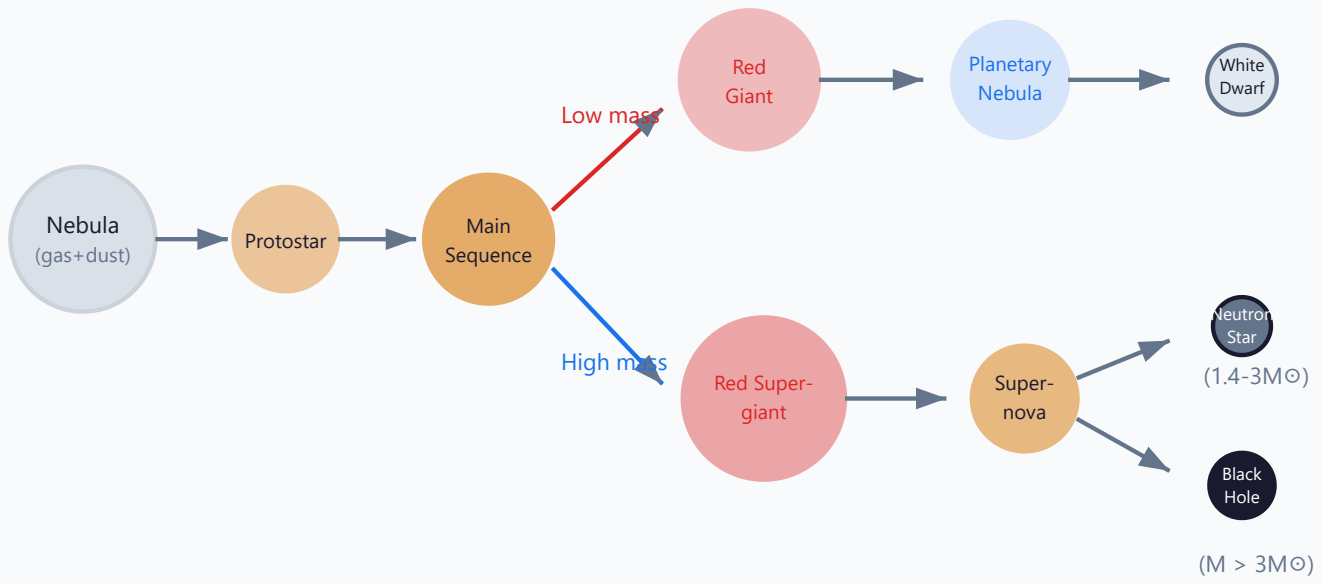
10.1 Stellar Evolution

Life Cycle of Stars

All stars begin in the same way but their ultimate fate depends on their **mass**. More massive stars live shorter, more dramatic lives.

Low-mass star (M < 8M⊙):
Nebula → Protostar → Main Sequence → Red Giant → Planetary Nebula → White Dwarf

High-mass star (M > 8M⊙):
Nebula → Protostar → Main Sequence → Red Supergiant → Supernova → Neutron Star or Black Hole



Main sequence: The longest stable phase of a star's life, during which the inward gravitational force is balanced by the outward radiation pressure from hydrogen fusion in the core. This equilibrium is called **hydrostatic equilibrium**.

Stages Explained

- **Nebula:** A large cloud of gas (mainly hydrogen) and dust. Gravity causes regions to collapse, triggered by events such as nearby supernova shockwaves.
- **Protostar:** A collapsing region that heats up due to gravitational potential energy being converted to kinetic (thermal) energy. Not yet hot enough for nuclear fusion. The protostar contracts over millions of years.
- **Main Sequence:** Core temperature reaches ~10^7 K, hydrogen fusion begins (pp chain or CNO cycle). Hydrostatic equilibrium is established. The Sun has been on the main sequence for ~4.6 billion years and will remain for another ~5 billion.
- **Red Giant / Supergiant:** Hydrogen fuel in the core is exhausted. The core contracts and heats up, outer layers expand and cool (hence red colour). Helium fusion begins in the core (triple-alpha process). In massive stars, successive heavier elements fuse up to iron-56.
- **Planetary Nebula:** The outer layers of a low-mass star are gently ejected, forming a beautiful expanding shell of gas illuminated by the hot core. Despite the name, planetary nebulae have nothing to do with planets.
- **White Dwarf:** The remnant core of a low-mass star. No fusion occurs; supported by **electron degeneracy pressure**. Mass limit: 1.4 $M_\odot$ (Chandrasekhar limit). Gradually cools over billions of years to become a black dwarf.
- **Supernova:** The catastrophic explosion of a massive star when iron accumulates in the core. Iron fusion is endothermic, so the core collapses in milliseconds, releasing ~10^{44} J. Elements heavier than iron are created in the supernova.
- **Neutron Star:** The ultra-dense remnant after a supernova (core mass 1.4–3 $M_\odot$). Supported by **neutron degeneracy pressure**. Density ~10^{17} kg m^{-3}. Pulsars are rotating neutron stars emitting beams of radiation.
- **Black Hole:** Forms if the remnant core mass exceeds ~3 $M_\odot$ (Tolman–Oppenheimer–Volkoff limit). Gravity is so strong that not even light can escape from within the event horizon.

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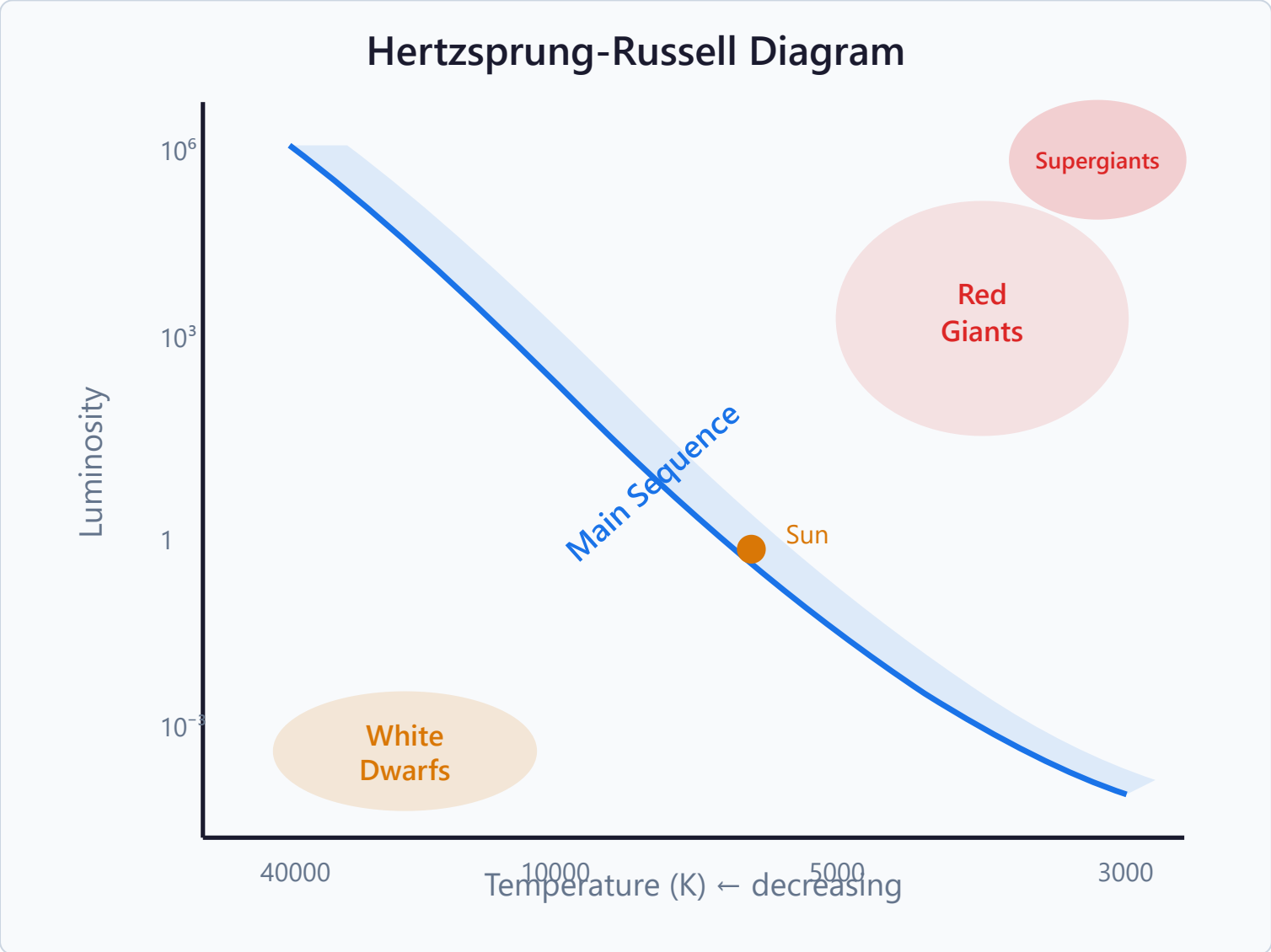
Exam Tip: A star spends about 90% of its life on the main sequence. The more massive the star, the shorter its main-sequence lifetime because it burns fuel much faster. A star of 10 M☉ lives only ~20 million years, compared to ~10 billion years for the Sun.

10.2 The Hertzsprung–Russell Diagram

Hertzsprung–Russell (HR) diagram: A plot of stellar luminosity (or absolute magnitude) against surface temperature (or spectral class). Temperature increases from right to left on the x-axis (an unusual convention).



Key Regions

Region	Position on HR Diagram	Characteristics
Main Sequence	Diagonal band from top-left to bottom-right	Hot, luminous blue stars at top; cool, dim red stars at bottom. Hydrogen fusion in core.
Red Giants	Upper right	Cool (3000–5000 K) but very luminous due to large radius. Helium fusion.
Red Supergiants	Top right	Even larger and more luminous. Fuse elements up to iron.
White Dwarfs	Bottom left	Hot (10,000–30,000 K) but very low luminosity due to tiny radius (~Earth-sized).

Exam Tip: The x-axis of the HR diagram has temperature *decreasing* from left to right (opposite to what you might expect). Always check the axis labels in exam questions. Also, both axes are usually logarithmic.

10.3 Luminosity and Brightness

Luminosity (L): The total power output of a star — the total energy radiated per second, measured in watts (W). The Sun's luminosity is $L_\odot = 3.85 \times 10^{26}$ W.

Apparent brightness (b): The intensity of light received at Earth from a star — the power per unit area at the observer, measured in W m^{-2}. This is what we actually measure with a telescope.

Apparent brightness depends on both the luminosity and the distance to the star (inverse square law):

$$b = L / (4\pi d^2)$$

where **d** is the distance from the star to the observer. The factor $4\pi d^2$ is the surface area of the sphere over which the star's light is spread.

Worked Example: The Sun has luminosity 3.85×10^{26} W. Calculate its apparent brightness at Earth ($d = 1.50 \times 10^{11}$ m).

$b = L / (4\pi d^2) = 3.85 \times 10^{26} / (4\pi \times (1.50 \times 10^{11})^2)$
$b = 3.85 \times 10^{26} / (2.83 \times 10^{23})$ = **1360 W m^{-2}**

This is the solar constant — the power per square metre received at Earth's orbital distance.

10.4 Wien's Displacement Law

Wien's displacement law: The peak wavelength of the radiation emitted by a black body is inversely proportional to its absolute (surface) temperature.

$$\lambda_{max} T = 2.898 \times 10^{-3} \text{ m K}$$

This means hotter stars emit radiation with a shorter peak wavelength (appearing bluer), while cooler stars have a longer peak wavelength (appearing redder).

Worked Example: A star has a surface temperature of 25 000 K. What is its peak wavelength and what colour does it appear?

$\lambda_{max} = 2.898 \times 10^{-3} / 25\,000 = 1.16 \times 10^{-7}$ m = **116 nm**

This is in the ultraviolet. The star would appear **blue-white** because although the peak is in the UV, it still emits plenty of visible light at the blue end of the spectrum.

10.5 Stefan–Boltzmann Law

Stefan–Boltzmann law: The total luminosity of a star (modelled as a black body) is proportional to the fourth power of its surface temperature and to its surface area.

$$L = 4\pi r^2 \sigma T^4$$

where:

- **L** = luminosity (W)
- **r** = radius of the star (m)
- **σ** = Stefan–Boltzmann constant = 5.67×10^{-8} W m^{-2} K^{-4}
- **T** = surface temperature (K)

The T^4 dependence means that a small increase in temperature leads to a dramatic increase in luminosity. Doubling the temperature increases the luminosity by a factor of $2^4 = 16$.

Worked Example: A red giant has surface temperature 3500 K and luminosity 400 L⊙. Estimate its radius in terms of solar radii. (Sun: T = 5800 K, R⊙ = 6.96×10^8 m)

Using $L = 4\pi r^2 \sigma T^4$, the ratio of luminosities gives:

$L_{giant}/L_\odot = (r_{giant}/R_\odot)^2 \times (T_{giant}/T_\odot)^4$

$400 = (r_{giant}/R_\odot)^2 \times (3500/5800)^4$

$400 = (r_{giant}/R_\odot)^2 \times 0.1327$

$(r_{giant}/R_\odot)^2 = 400/0.1327 = 3014$

$r_{giant}/R_\odot = \sqrt{3014} =$ **55 R⊙**

The red giant is 55 times larger than the Sun, which explains why it appears in the upper right of the HR diagram (cool but very luminous due to its large surface area).

Exam Tip: You can combine Wien's law and Stefan–Boltzmann law to determine both the temperature and radius of a star from its spectrum and luminosity. First use Wien's law with the peak wavelength to find T, then use Stefan-Boltzmann to find r.

10.6 Standard Candles

Standard candle: An astronomical object whose absolute luminosity is known (or can be determined independently), allowing its distance to be calculated from its observed apparent brightness using $b = L/(4\pi d^2)$.

Cepheid Variable Stars

Cepheid variables are pulsating stars whose luminosity varies periodically. There is a well-established **period-luminosity relationship**: brighter Cepheids pulsate with longer periods. By measuring the period, we determine the luminosity, and then using the apparent brightness, we can calculate the distance.

Cepheids are crucial because they allowed Edwin Hubble to prove that other galaxies exist beyond the Milky Way (1924) and to establish the distance scale of the universe.

Type Ia Supernovae

Type Ia supernovae occur in binary star systems when a white dwarf accretes matter from a companion until it reaches the **Chandrasekhar limit** ($\sim 1.4\ M_\odot$) and explodes. Since they all explode at approximately the same mass, they have very similar peak luminosities, making them excellent standard candles for measuring very large (cosmological) distances.

Type Ia supernovae can be seen across billions of light-years. They were used in 1998 to discover that the expansion of the universe is **accelerating**, leading to the concept of dark energy (Nobel Prize 2011 to Perlmutter, Schmidt, and Riess).

10.7 The Doppler Effect

Doppler effect: The change in observed frequency (or wavelength) of a wave due to relative motion between the source and the observer.

For electromagnetic radiation (when $v << c$):

$$\Delta\lambda / \lambda = \Delta f / f \approx v / c$$

where $\Delta\lambda$ = change in wavelength (observed − emitted), λ = emitted wavelength, v = speed of the source relative to the observer, c = speed of light.

Redshift and Blueshift

- **Redshift (z > 0):** Source is moving away — observed wavelength is longer (shifted towards the red end). All distant galaxies are redshifted.
- **Blueshift (z < 0):** Source is moving towards observer — observed wavelength is shorter (shifted towards the blue end). The Andromeda galaxy is blueshifted.

$$z = \Delta\lambda / \lambda = v / c$$

Worked Example: A hydrogen spectral line (rest wavelength 656.3 nm) from a distant galaxy is observed at 670.4 nm. Calculate the galaxy's recessional velocity.

$\Delta\lambda$ = 670.4 − 656.3 = 14.1 nm
z = $\Delta\lambda/\lambda$ = 14.1/656.3 = 0.0215
v = zc = 0.0215 × 3.00 × 10^8 = **6.44×10^6 m s^{-1}** ≈ 6400 km s^{-1}

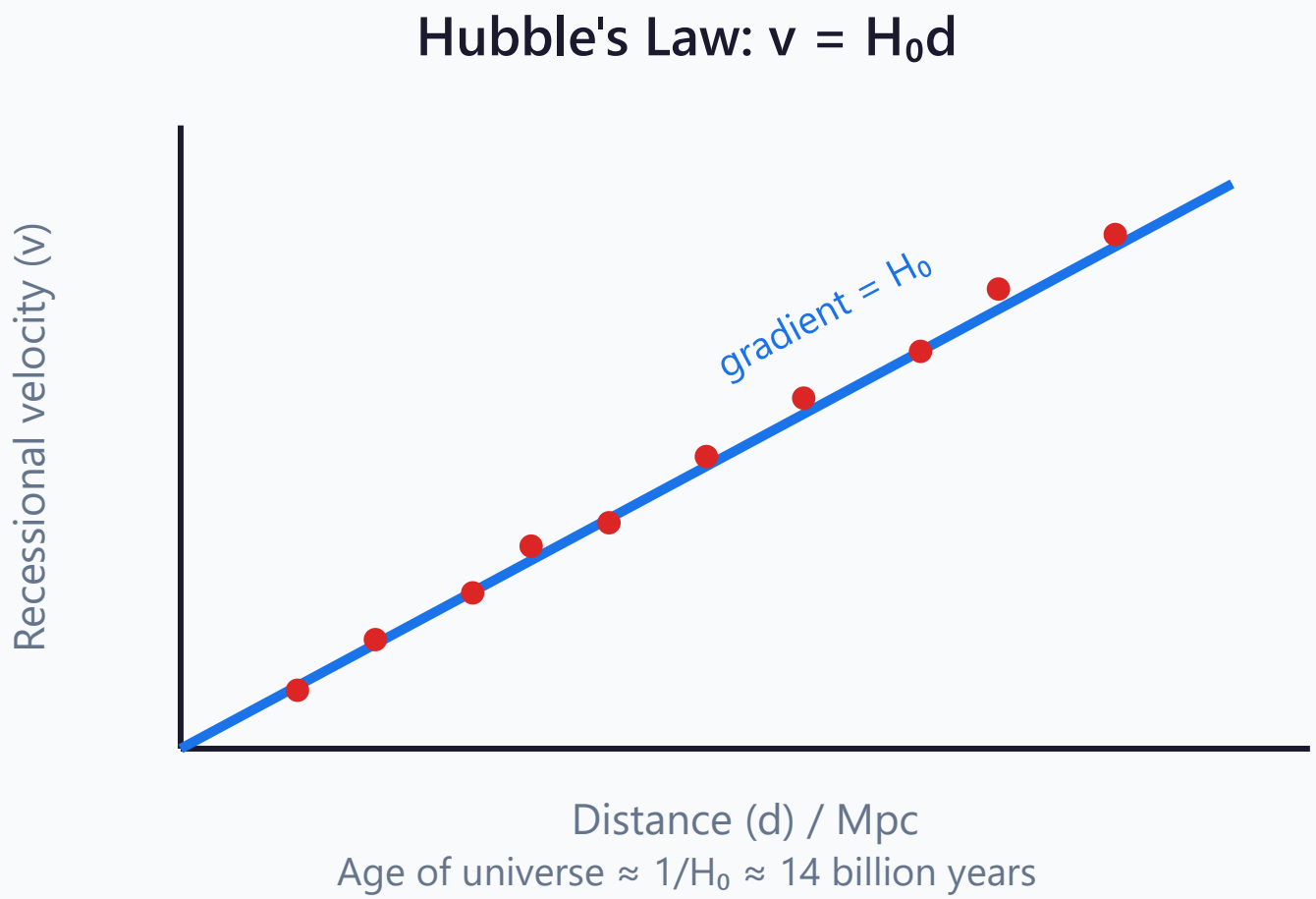
The galaxy is moving away at about 2% of the speed of light.

10.8 Hubble's Law

Hubble's law: The recessional velocity of a distant galaxy is directly proportional to its distance from us.

$$v = H_0 d$$

where v = recessional velocity (km s^{-1} or m s^{-1}), H_0 = Hubble constant (approximately 67–73 km s^{-1} Mpc^{-1}), d = distance to the galaxy.



Hubble's law implies that the universe is **expanding** — all distant galaxies are moving away from us, and the further away they are, the faster they recede. This does not mean we are at the centre; every point in the universe sees the same pattern (like dots on an inflating balloon).

Age of the Universe

If the expansion rate has been approximately constant, the age of the universe can be estimated as:

$$t \approx 1 / H_0$$

Worked Example: Estimate the age of the universe using $H_0 = 70$ km s^{-1} Mpc^{-1}.

First convert H_0 to SI units (s^{-1}):

1 Mpc = 3.09×10^{22} m, and 1 km = 10^3 m

$H_0 = 70 \times 10^3 / (3.09 \times 10^{22}) = 2.27 \times 10^{-18}$ s^{-1}

$t = 1/H_0 = 1 / (2.27 \times 10^{-18}) = 4.41 \times 10^{17}$ s

Converting to years: $4.41 \times 10^{17} / (3.15 \times 10^{7})$ = **1.4×10^{10} years ≈ 14 billion years**

Exam Tip: Be careful with units when calculating the age. Convert H_0 to s^{-1} by converting Mpc to metres (1 Mpc = 3.09×10^{22} m) and km to m. This unit conversion is commonly tested and frequently causes errors.

10.9 Evidence for the Big Bang

The **Big Bang theory** states that the universe began from an extremely hot, dense singularity approximately 13.8 billion years ago and has been expanding and cooling ever since.

Key Evidence

Evidence	Explanation
Cosmic Microwave Background (CMB) Radiation	Uniform microwave radiation (peak wavelength ~1 mm, T ≈ 2.7 K) detected in all directions. This is the cooled remnant of the radiation from ~380,000 years after the Big Bang when the universe became transparent (recombination epoch). Predicted by Gamow (1948), discovered by Penzias and Wilson (1965, Nobel Prize).
Expansion of the Universe	Hubble's observations show all distant galaxies are receding (redshifted), implying the universe is expanding — consistent with expansion from an initial singularity. Running the expansion backwards leads to a single point in space-time.
Hydrogen/Helium Abundance	The observed ratio (~75% hydrogen, ~25% helium by mass) matches predictions of Big Bang nucleosynthesis — the formation of light elements in the first few minutes after the Big Bang when the temperature was ~10^9 K.

Exam Tip: The CMB is the strongest single piece of evidence for the Big Bang. Its near-perfect black-body spectrum at 2.7 K is exactly what the model predicts for radiation that has been cooling for

13.8 billion years. Tiny fluctuations (~1 part in 100,000) in the CMB correspond to the density variations that seeded the formation of galaxies.

Dark Matter and Dark Energy

Observations suggest that ordinary (baryonic) matter makes up only about 5% of the total energy content of the universe:

- **Dark matter (~27%):** Invisible matter detected through its gravitational effects on galaxy rotation curves and gravitational lensing. Galaxies rotate too fast to be held together by visible matter alone.
- **Dark energy (~68%):** A mysterious form of energy causing the expansion of the universe to accelerate. Discovered in 1998 through observations of Type Ia supernovae at great distances.

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